Sprint Ceremony Document: Sprint 3

Project Name: ALT-F4
Sprint Window: October 25 – October 31, 2026
Overall Completion: 71/168 * 100 = **42.26%**
Sprint 3 Completion: 24/24 * 100 = **100%**

Sprint Planning & Progress Update

Completed Tasks

1. Clinic Slot API & Scheduling Validation (CQ-S3-3.1)

- Slot Listing by Date Range: Implemented API endpoints to retrieve clinic appointment slots within specified date ranges.
- Slot Creation & Updates: Implemented endpoints for creating and updating clinic slots, with validation for required dates, times, and capacity.
- Overlapping Schedule Prevention: Prevented overlapping clinic slots on the same date to maintain schedule integrity.
- Safe Slot Deletion: Prevented deletion of slots with active bookings to protect existing patient appointments.

2. Staff Schedule Management Interface (CQ-S3-3.2)

- Dynamic Schedule Dashboard: Developed a staff schedule page displaying appointment dates, times, total capacity, remaining capacity, and slot statuses (open, full, or closed).
- Slot Management Forms: Implemented forms for adding and editing slots, with validation for required fields, chronological time order, past dates, and capacity restrictions.
- API Integration: Connected the schedule interface to the slot APIs for listing, creating, updating, and deleting clinic slots.
- Conflict Handling & Feedback: Added validation messages, conflict notifications, loading and empty states, error handling, and automatic schedule refreshes after successful operations.
- Schedule Integrity & Safety: Added delete confirmation dialogs, locked date and time fields for slots with existing bookings, and corrected date-saving issues that caused slots to be saved one day early.

3. Patient Availability & Booking Interface (CQ-S3-3.3)

- Live Slot Discovery: Connected the patient booking interface to clinic slots created by staff, automatically hiding full or unavailable slots.
- Dynamic Appointment Dashboard: Implemented appointment loading with loading indicators, empty states, and error-retry functionality.

- Booking Submission & Confirmation: Enabled patients to select available slots, provide consultation details, and submit appointments with success and conflict feedback.
- Appointment Cancellation: Implemented patient cancellation with confirmation prompts and automatic restoration of available slot capacity.

4. Appointment Management APIs (CQ-S3-3.4)

- Appointment Creation & Validation: Implemented appointment creation with server-side slot availability checks, rejecting bookings for full or unavailable slots.
- Appointment Listing & Details: Established appointment retrieval behavior and confirmed the API contract against the defined user stories.
- Cancellation & Rescheduling: Implemented appointment cancellation and rescheduling workflows with corresponding slot capacity updates.
- Capacity Integrity: Ensured that appointment operations maintain accurate availability and prevent invalid booking transactions.

5. Frontend Integration, Automated Testing & Quality Assurance (CQ-S3-3.5 & CQ-S3-3.6)

- End-to-End Booking Integration: Connected `BookingsComponent` and `PatientDashboardComponent` to the live `SlotsService` and `AppointmentsService`, replacing placeholder data with backend-driven appointment information.
- Dynamic Capacity Synchronization: Verified that remaining capacity updates correctly after booking, rescheduling, and cancellation, reflecting the number of active `BOOKED` appointments.
- Conflict & Duplicate Booking Protection: Implemented HTTP `409 Conflict` handling and automatic schedule refreshes to reconcile outdated availability and prevent conflicting submissions.
- Automated Workflow Testing: Developed a multi-role end-to-end smoke test covering staff slot creation, patient booking, capacity changes, duplicate booking rejection, rescheduling, cancellation, and safe slot deletion.
- Regression Testing & Verification: Fixed booking and patient dashboard unit test regressions, verified date/time and slot integrity rules, and achieved 110 passing web tests.
- Defect Tracking & Demonstration Preparation: Tested booking capacity and appointment management flows, logged defects, verified fixes, and prepared the system for demonstration.

Planned Tasks for Next Sprint (Sprint 4):

Sprint 4

Task ID	Task Title	Description
CQ-S4-4.1	Finalize virtual queue rules and data fields	Define active/closed queue behavior, ticket numbering, position/wait estimate, priority, timestamps, tracking token and duplicate-entry rules.
CQ-S4-4.2	Implement walk-in queue entry	Implement validated walk-in registration and return tracking details such as ticket number, position, wait estimate, status and tracking token when needed.
CQ-S4-4.3	Implement queue status retrieval and integrity rules	Provide queue status metrics and ensure invalid IDs, closed queues, duplicate entries and unauthorized tracking-token access are handled safely.
CQ-S4-4.4	Build the patient virtual queue screen	Build the patient queue view showing ticket, position, estimated wait, current serving/status and updated state.
CQ-S4-4.5	Integrate and test the virtual queue	Connect the queue UI to entry/status APIs and test numbering, wait estimates, invalid input, closed queues, duplicates and access protection.
CQ-S4-4.6	Finalize working components	Collect asynchronous team updates and produce a short progress recording showing working screens, queue behavior, blockers and next steps.

Sprint Retrospective

What Went Well:

- Clinic Slot API & Scheduling Integrity (CQ-S3-3.1): Implemented slot listing, creation, updating, and deletion with date, time, capacity, and overlap validation. Prevented deletion of slots with active bookings to preserve appointment integrity.
- Staff Schedule Management & UI/API Integration (CQ-S3-3.2): Delivered a functional staff scheduling interface connected to the backend, featuring dynamic capacity information, slot status indicators, validated forms, confirmation dialogs, and conflict handling.
- Patient Booking & Appointment Management (CQ-S3-3.3 & CQ-S3-3.4): Integrated live slot discovery with appointment creation, listing, cancellation, and rescheduling. Patients can select available slots, submit consultation details, and receive booking feedback.
- Dynamic Capacity Synchronization (CQ-S3-3.4 & CQ-S3-3.5): Established consistent capacity updates across booking, rescheduling, and cancellation. Full or unavailable

slots are rejected, while cancellation restores capacity and rescheduling transfers capacity between slots.
- Automated Testing & Quality Assurance (CQ-S3-3.5 & CQ-S3-3.6): Developed end-to-end smoke tests covering staff and patient workflows, verified scheduling and booking constraints, resolved frontend unit test regressions, and achieved 110 passing web tests.

What Could Be Improved:

- Date and Time Consistency Across the Application:
    - Challenge: Slot dates were being saved one day earlier than intended, creating potential discrepancies between the staff schedule and patient booking interface.
    - Resolution: Corrected the date-handling issue and added date and time validation, including chronological time checks and restrictions against past dates.
    - Future Improvement: Expand testing across different time zones, date boundaries, and browser environments to prevent similar scheduling issues.
- Frontend and Backend Validation Alignment:
    - Challenge: Booking and schedule-management operations require consistent validation between the frontend forms and backend APIs, particularly for overlapping slots, capacity restrictions, and unavailable appointments.
    - Resolution: Connected the interfaces to the slot and appointment APIs, added validation and conflict messages, and implemented server-side checks for slot availability.
    - Future Improvement: Maintain a centralized set of documented business rules and verify that frontend validation consistently matches backend enforcement.
- Real-Time Capacity and Booking Conflict Handling:
    - Challenge: Available capacity may change when multiple users attempt to book the same slot or when a user's displayed schedule becomes outdated.
    - Resolution: Implemented HTTP `409 Conflict` handling, automatic schedule refreshes, and capacity recalculation following booking, rescheduling, and cancellation.
    - Future Improvement: Conduct additional concurrent booking tests to verify that simultaneous requests cannot exceed slot capacity and that failed operations do not leave inconsistent appointment or capacity data.
- Comprehensive Test Coverage and Reporting:
    - Challenge: Although the 110 web tests passed, scheduling and booking involve several interconnected workflows, including duplicate submissions, full slots, overlapping schedules, and appointment cancellation and rescheduling.
    - Resolution: Added multi-role end-to-end smoke testing and verified the major booking and capacity-management workflows.

    - Future Improvement: Expand automated regression coverage for edge cases, authorization failures, simultaneous requests, and unexpected API errors. Record individual test results and outstanding defects more explicitly.
- Cross-Role Workflow Verification:
    - Challenge: Staff and patient interfaces depend on the same appointment and capacity data, so changes made by one role must be reflected correctly for the other.
    - Resolution: Verified that staff-created slots appear in patient booking, booking updates remaining capacity, rescheduling transfers capacity, and cancellation restores availability.
    - Future Improvement: Continue validating complete staff-to-patient workflows to ensure that interface updates, database state, and appointment statuses remain consistent.

What Needs to Be Done:

- Perform additional end-to-end testing of appointment booking, rescheduling, cancellation, duplicate submissions, full slots, and overlapping schedules.
- Verify backend transaction safety and concurrency controls to prevent overbooking when multiple patients attempt to reserve the same slot.
- Review role-based authorization and appointment ownership to ensure that patients can access and manage only their permitted appointments.
- Expand regression testing for date/time boundaries, capacity restrictions, API failures, and synchronization between staff and patient dashboards.
- Review and resolve outstanding defects identified during testing, documenting reproduction steps, fixes, and verification results.
- Conduct a complete system demonstration and gather feedback from intended users to identify usability issues and additional requirements for the next sprint.
- Reassess the sprint backlog and prioritize remaining improvements based on testing results, system stability, and user feedback.

Sprint 3 Task Reports
Sprint: CQ-S3-3.1

Member: John Albert D. Masangkay / Matthew Ryan R. Sabino
Role: Technical Lead / Project Manager
Task: Implement clinic slot APIs and validation to ensure that slot listing, creation, updates and deletion follow the required scheduling and capacity rules.

Done:

- Implemented slot listing by date range.

- Implemented create and update slot endpoints with required input validation for date, time and capacity.
- Prevented overlapping slots on the same date.
- Prevented deleting of appointments when there is active booking.

Sprint: CQ-S3-3.2

Member: Jereign B. Lim
Role: UI/UX Designer
Task: Build the staff schedule-management interface for listing, adding, editing and deleting clinic slots, connected to the slot API with validation and conflict handling.

Done:

- Built the staff schedule page with a slot list showing date, time, capacity, remaining capacity and open, full or closed status, with loading, empty and error states.
- Built the add and edit slot form with validation (required fields, end after start, no past dates, capacity not below booked slots) and locked date and time when slots are booked.
- Connected list, create, update and delete to the slot API, showed validation and conflict messages, and refreshed the list after success.
- Added a delete confirmation dialog.
- Fixed slot dates being saved one day early and added overlap and time-order checks in the API.

Sprint: CQ-S3-3.3

Member: John Alec L. Resos
Role: Tester
Task: Build patient availability and booking interface

Done:

- Connected the patient booking view to live clinic slots created by staff, automatically hiding full or unavailable slots.
- Added dynamic appointment loading on the patient dashboard with loading, empty, and error retry states.
- Allowed patients to select an available slot, enter consultation details, and confirm bookings with real-time success and conflict feedback.
- Enabled permitted patient actions, including appointment cancellation with a confirmation prompt that restores slot capacity.

Sprint: CQ-S3-3.4

Member: John Albert D. Masangkay / Matthew Ryan R. Sabino / John Raphael Reganit
Role: Technical Lead / Project Manager / Tester

Task: Implement create, list and detail behavior to ensure that appointments are created correctly and only permitted records are returned.

Done:

- Implemented appointment creation with slot availability validation.
- Rejected bookings for full or unavailable slots.
- Implement cancellation/reschedule and capacity updates
- Confirm appointment API contract from user stories

Sprint: CQ-S3-3.5

Member: Matthew Ryan Sabino
Role: Project Manager
Task: Integrate booking UI, appointment APIs and staff slots.

Done:

- **Backend Integration:** Connected the patient booking interface and dashboard to live appointment and slot APIs using `BookingsComponent`, `PatientDashboardComponent`, `SlotsService`, and `AppointmentsService`.
- **Dynamic Slot Availability:** Implemented real-time slot availability and capacity synchronization across booking, rescheduling, and cancellation, ensuring remaining capacity accurately reflects active `BOOKED` appointments.
- **Booking Lifecycle and Error Handling:** Implemented booking submission, loading states, confirmation receipts, cancellation, and rescheduling, with `409 Conflict` handling and automatic schedule refreshes to prevent duplicate bookings and resolve outdated availability.
- **Automated Testing and Verification:** Developed a multi-role end-to-end smoke test covering staff slot creation and patient booking workflows, including capacity validation, conflict handling, rescheduling, cancellation, and safe slot deletion.
- **Test Reliability and Validation:** Fixed regressions in the booking and patient dashboard unit tests, verified database capacity calculations and appointment status transitions, and achieved **110 passing web tests**.

Sprint: CQ-S3-3.6

Member: John Alec L. Resos / John Raphael Reganit / Matthew Ryan Sabino
Role: Tester / Project Manager
Task: Test validation, full/unbookable slots, overlap/capacity constraints, duplicate submissions and cancel/reschedule paths.

Done:

- Test schedule and appointment rules
- Test date/time and slot integrity rules
- Log defects, verify fixes and prepare demo
- Duplicate Submissions
- Test booking capacity and management flows

S3-3.6 TEST MATRIX

CLINIQUE SYSTEM TEST MATRIX - SCHEDULING & APPOINTMENTS

This document covers the booking workflow only.

CQ-S3-3.6.01 - Schedule & appointment rules

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
SLOT-01	**Integration**	Create a slot with an invalid calendar date	clinicDate 2026-13-45	Validation error shown, no slot created	**Success**
SLOT-02	**Integration**	Create a slot with a malformed date format	clinicDate 11-20-2026	Validation error shown, no slot created	**Success**
SLOT-03	**Integration**	Create a slot with an out-of-range time	startTime 25:00	Validation error shown, no slot created	**Success**

SLOT-04	**Integration**	Create a slot with end time before start	Start 15:00, end 14:00	Validation error shown, no slot created	**Success**
SLOT-05	**Integration**	Create a slot with end time equal to start	Start 15:00, end 15:00	Validation error shown, no slot created	**Success**
SLOT-06	**Integration**	Create a slot with capacity below minimum	capacity 0	Validation error shown, no slot created	**Success**
SLOT-07	**Integration**	List slots using an inverted date range	to date earlier than from date	Validation error shown, no list returned	**Success**
SLOT-08	**Integration**	Create a slot dated in the past	clinicDate 2020-01-01	Slot rejected, past dates not permitted	**Fail**
SLOT-09	**Integration**	Create a slot overlapping an existing slot	09:15-09:45 over 09:00-09:30	Slot rejected, conflict reported	**Success**

SLOT-10	**Integration**	Create a slot touching an existing boundary	09:30-10:00 after 09:00-09:30	Slot created, ranges only touch	**Success**
SLOT-11	**Integration**	Update a slot without any field	Empty JSON object	Validation error shown, slot unchanged	**Success**
SLOT-12	**Integration**	Patient attempts to create a clinic slot	Authenticated patient token	Access denied, staff and admin only	**Success**
SLOT-13	**Integration**	Reduce a slot capacity below booked seats	One booked seat, set to 0	Reduction rejected, capacity unchanged	**Success**
SLOT-14	**Integration**	Delete a slot with an active appointment	Slot with one BOOKED appointment	Deletion rejected, slot retained	**Success**
SLOT-15	**Integration**	Delete a slot with no active appointments	Slot with no bookings	Slot deleted successfully	**Success**

CQ-S3-3.6.02 - Appointment & booking management

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
BOOK-01	**E2E**	Patient books an available appointment slot	Valid patient account, date/time slot, req	Appointment is created successfully and a booking confirmation	**Success**
BOOK-02	**Unit**	System validates required booking fields	Missing patient info, date or details	Validation errors are displayed and invalid data is rejected	**Success**
BOOK-03	**Integration**	System updates slot availability after successful booking	System updates slot availability	Slot availability is updated in DB and reflected in interface	**Success**
BOOK-04	**Unit&Integ.**	Patient attempts to book a fully booked slot	Appointment slot with no remaining capacity	Booking is rejected and system informs slot is unavailable	**Success**

BOOK-05	**Integration**	Two patients attempt to book final slot simultaneously	Two patient accounts and one slot	Only one booking succeeds, no overbooking	**Fail**
BOOK-06	**Unit**	System validates appointment date and time	Past date, invalid time, or outside allowed period	Invalid schedules are rejected according to scheduling rules	**Fail**
BOOK-07	**E2E**	Patient views existing appointments	Authenticated patient with bookings	Only authenticated patient's appointments displayed	**Success**
BOOK-08	**E2E**	Patient cancels an existing appointment	Authenticated patient with active booking	Status updated to cancelled and reflected in records	**Success**
BOOK-09	**Integration**	System releases a slot after cancellation	Active booking occupying slot	Cancelled booking no longer occupies slot	**Success**
BOOK-10	**E2E**	Patient reschedules	Active booking	Original booking updated or replaced,	**Success**

		existing appointment	and new slot	new schedule recorded	
BOOK-11	**E2E**	Patient attempts to reschedule to unavailable slot	Active booking and fully booked dest slot	Rescheduling rejected, original appointment unchanged	**Success**
BOOK-12	**Unit & Integ.**	System prevents duplicate submissions	Same patient, details, repeated request	Duplicates do not create unintended bookings	**Success**
BOOK-13	**Unit**	System validates booking capacity	Slot capacity, current bookings, new req	Accepts within capacity, rejects exceeding capacity	**Success**
BOOK-14	**E2E**	Walk-in patient requests an appointment at clinic	Patient details, requested service, slot	Registered successfully and added to active queue with online bookings	**Fail**
BOOK-15	**Integration**	Patient books the same slot twice	Same patient, same slot,	Duplicate booking rejected, one	**Success**

			second request	appointment remains	
BOOK-16	**Integration**	Patient books a second slot at the same date and time	Two slots sharing a date and start time	Booking rejected, date and time already held	**Success**
BOOK-17	**Integration**	Patient books a slot closed by staff	Slot marked as not bookable	Booking rejected, slot reported unavailable	**Success**
BOOK-18	**E2E**	Patient views a single appointment detail	Own appointment identifier	Appointment and slot details returned	**Success**
BOOK-19	**Integration**	Patient views an appointment that does not exist	Appointment identifier 999999	System reports appointment not found	**Success**
BOOK-20	**Integration**	Patient cancels an already cancelled appointment	Appointment in CANCELLED status	Cancellation rejected, status unchanged	**Success**

BOOK-21	**E2E**	Clinic staff views scheduled appointments	Authenticated staff account with records	Staff views appointments within authorized scope	**Success**
BOOK-22	**Integration**	Patient reschedules a cancelled appointment	Appointment in CANCELLED status	Reschedule rejected, status unchanged	**Success**
BOOK-23	**Integration**	Patient reschedules to the slot already held	Same slot identifier as current booking	Reschedule rejected, appointment unchanged	**Success**
BOOK-24	**Integration**	Patient reschedules to a slot that does not exist	Slot identifier 999999	Reschedule rejected, appointment unchanged	**Success**
BOOK-25	**Integration**	Patient views another patient's appointment	Foreign appointment identifier	Access denied, appointment not disclosed	**Success**
BOOK-26	**Integration**	Patient reschedules another patient's appointment	Foreign appointment identifier	Access denied, appointment unchanged	**Success**

BOOK-27	**Integration**	Patient cancels another patient's appointment	Foreign appointment identifier	Access denied, appointment unchanged	**Success**

CQ-S3-3.6.03 - Defects logged

DEF-01 - Past clinic dates are accepted

Creating a slot with `clinicDate` set to `2020-01-01` returned `200` and created the slot.

Field	Value
Test	SLOT-08
Owner	Slots module

DEF-02 - Past-dated slots can still be booked

A patient booked an appointment into the slot dated `2020-01-01` and received a successful response. Even if slot creation is later restricted, existing past-dated slots remain bookable because the booking path performs no date comparison.

Field	Value
Test	BOOK-06

Owner	Appointments module

DEF-03 - Concurrent bookings overbook a slot

Two races were reproduced against real records.

Six identical requests issued at the same moment against a capacity-5 slot produced two successful bookings. Two different patients booking a capacity-1 slot at the same moment both received a successful response, leaving two `BOOKED` appointments against a slot holding one seat. Reading the slot afterwards reported a remaining capacity of zero rather than a negative value, so the API does not reveal that it is oversold.

Field	**Value**
Test	BOOK-05
Owner	Appointments module

DEF-04 - Walk-in registration does not create an appointment

The walk-in endpoint registers a queue entry and returns a ticket, but no appointment record is created, so the patient is queued without a booked schedule.

Field	**Value**
Test	BOOK-14
Owner	Appointments and queue modules

CLINIQUE SYSTEM TEST MATRIX:

This document outlines the functional and system data testing scenarios for the CliniQueue system to evaluate its overall functionality, reliability, performance, and compliance with the defined system requirements.

AUTHENTICATION & ROLE AUTHENTICATION

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
AUTH-01	**Integration**	Admin sends a registration link to a valid DLSU-D account	Valid DLSU-D email	Registration link is sent to the intended email	**Pending**
AUTH-02	**E2W**	Admin attempts to send a registration link to a non-DLSU-D email	Personal email	Registration is rejected	**Pending**
AUTH-03	**E2E**	User opens registration link	Registration link & form	Registration page opens successfully	**Pending**
AUTH-04	**E2E**	User registers with valid information	DLSU-D email, password, and req. fields	Account is created successfully	**Pending**

AUTH-05	**Unit**	User attempts registration with an expired link	Expired registration link	Registration is denied; user must request a new link	**Pending**
AUTH-06	**Unit**	User attempts to reuse a registration link	Link has already been used	Duplicate registration is prevented	**Pending**
AUTH-07	**Unit**	User submits incomplete registration details	Missing required fields	Validation errors are displayed	**Success**
AUTH-08	**E2E**	User submits a weak password	Password that does not meet requirements	Validation errors are displayed	**Success**
AUTH-09	**E2E**	User signs in with valid credentials	Registered account and correct password	User is authenticated and redirected to appropriate dashboard	**Success**
AUTH-10	**E2E**	User attempts to sign in before	User	Access is denied	**Success**

		completing registration			
AUTH-11	**E2E**	User attempts to access a protected page without signing in	No active authenticated session	User is redirected to sign-in or denied access	**Success**
AUTH-12	**E2E**	User attempts to access another role's restricted page	Authenticated user without required role	Access is denied	**Success**
AUTH-13	**E2E**	User signs out and revisits a protected page	Previously authenticated user who has signed out	Protected content is no longer accessible without signing in again	**Success**
AUTH-14	**Integration**	User requests a password reset using a registered email	Registered email	Password reset instructions are sent to the registered email	**Fail**
AUTH-15	**E2E**	User opens a valid password reset link	Valid, unused reset link	Password reset page opens	**Pending**

				successfully	
AUTH-16	**Unit**	User opens an expired password reset link	Expired reset link	Password reset is denied; user is prompted to request new link	**Pending**
AUTH-17	**Unit**	User reuses a password reset link	Previously used reset link	Password reset is denied	**Pending**
AUTH-18	**E2E**	User resets password with valid input	Valid reset link and password reqs	Password is updated successfully	**Pending**
AUTH-19	**Unit**	User submits a weak or invalid new password	Password that does not meet requirements	Validation error is displayed and password remains same	**Pending**
AUTH-20	**Unit**	User submits mismatched new passwords	Password and confirmation differ	Validation error is displayed and password is not changed	**Pending**

AUTH-21	**E2E**	User signs in using the new password	New password	Login succeeds with the new password	**Pending**
AUTH-22	**E2E**	User signs in using the old password	Old password	Login is rejected using the old password	**Pending**
AUTH-23	**Integration**	User requests multiple password reset links	Several reset requests submitted for acc	Reset tokens follow system's defined policy; only valid ones can be used	**Fail**

APPOINTMENT & BOOKING MANAGEMENT

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
BOOK-01	**E2E**	Patient books an available appointment slot	Valid patient account, date/time slot, req	Appointment is created successfully and a booking confirmation	**Success**
BOOK-02	**Unit**	System validates required	Missing patient	Validation errors are displayed and	**Success**

		booking fields	info, date or details	invalid data is rejected	
BOOK-03	**Integration**	System updates slot availability after successful booking	System updates slot availability	Slot availability is updated in DB and reflected in interface	**Success**
BOOK-04	**Unit&Integ.**	Patient attempts to book a fully booked slot	Appointment slot with no remaining capacity	Booking is rejected and system informs slot is unavailable	**Success**
BOOK-05	**Integration**	Two patients attempt to book final slot simultaneously	Two patient accounts and one slot	Only one booking succeeds, no overbooking	**Fail**
BOOK-06	**Unit**	System validates appointment date and time	Past date, invalid time, or outside allowed period	Invalid schedules are rejected according to scheduling rules	**Fail**
BOOK-07	**E2E**	Patient views existing appointments	Authenticated patient with bookings	Only authenticated patient's appointments displayed	**Success**

BOOK-08	**E2E**	Patient cancels an existing appointment	Authenticated patient with active booking	Status updated to cancelled and reflected in records	**Success**
BOOK-09	**Integration**	System releases a slot after cancellation	Active booking occupying slot	Cancelled booking no longer occupies slot	**Success**
BOOK-10	**E2E**	Patient reschedules existing appointment	Active booking and new slot	Original booking updated or replaced, new schedule recorded	**Success**
BOOK-11	**E2E**	Patient attempts to reschedule to unavailable slot	Active booking and fully booked dest slot	Rescheduling rejected, original appointment unchanged	**Success**
BOOK-12	**Unit & Integ.**	System prevents duplicate submissions	Same patient, details, repeated request	Duplicates do not create unintended bookings	**Success**
BOOK-13	**Unit**	System validates booking capacity	Slot capacity, current	Accepts within capacity, rejects	**Success**

			bookings, new req	exceeding capacity	
BOOK-14	**E2E**	Walk-in patient requests an appointment at clinic	Patient details, requested service, slot	Registered successfully and added to active queue with online bookings	**Fail**
BOOK-15	**Integration**	Patient books the same slot twice	Same patient, same slot, second request	Duplicate booking rejected, one appointment remains	**Success**
BOOK-16	**Integration**	Patient books a second slot at the same date and time	Two slots sharing a date and start time	Booking rejected, date and time already held	**Success**
BOOK-17	**Integration**	Patient books a slot closed by staff	Slot marked as not bookable	Booking rejected, slot reported unavailable	**Success**
BOOK-18	**E2E**	Patient views a single appointment detail	Own appointment identifier	Appointment and slot details returned	**Success**
BOOK-19	**Integration**	Patient views an appointme	Appointment	System reports	**Success**

		nt that does not exist	identifier 999999	appointment not found	
BOOK-20	**Integration**	Patient cancels an already cancelled appointment	Appointment in CANCELLED status	Cancellation rejected, status unchanged	**Success**
BOOK-21	**E2E**	Clinic staff views scheduled appointments	Authenticated staff account with records	Staff views appointments within authorized scope	**Success**
BOOK-22	**Integration**	Patient reschedules a cancelled appointment	Appointment in CANCELLED status	Reschedule rejected, status unchanged	**Success**
BOOK-23	**Integration**	Patient reschedules to the slot already held	Same slot identifier as current booking	Reschedule rejected, appointment unchanged	**Success**
BOOK-24	**Integration**	Patient reschedules to a slot that does not exist	Slot identifier 999999	Reschedule rejected, appointment unchanged	**Success**

BOOK-25	**Integration**	Patient views another patient's appointment	Foreign appointment identifier	Access denied, appointment not disclosed	**Success**
BOOK-26	**Integration**	Patient reschedules another patient's appointment	Foreign appointment identifier	Access denied, appointment unchanged	**Success**
BOOK-27	**Integration**	Patient cancels another patient's appointment	Foreign appointment identifier	Access denied, appointment unchanged	**Success**

SCHEDULE & APPOINTMENT RULES

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
SLOT-01	**Integration**	Create a slot with an invalid calendar date	clinicDate 2026-13-45	Validation error shown, no slot created	**Success**
SLOT-02	**Integration**	Create a slot with a malforme	clinicDate 11-20-2026	Validation error shown,	**Success**

		d date format		no slot created	
SLOT-03	**Integration**	Create a slot with an out-of-range time	startTime 25:00	Validation error shown, no slot created	**Success**
SLOT-04	**Integration**	Create a slot with end time before start	Start 15:00, end 14:00	Validation error shown, no slot created	**Success**
SLOT-05	**Integration**	Create a slot with end time equal to start	Start 15:00, end 15:00	Validation error shown, no slot created	**Success**
SLOT-06	**Integration**	Create a slot with capacity below minimum	capacity 0	Validation error shown, no slot created	**Success**
SLOT-07	**Integration**	List slots using an inverted date range	to date earlier than from date	Validation error shown, no list returned	**Success**
SLOT-08	**Integration**	Create a slot dated in the past	clinicDate 2020-01-01	Slot rejected, past dates not permitted	**Fail**

SLOT-09	**Integration**	Create a slot overlapping an existing slot	09:15-09:45 over 09:00-09:30	Slot rejected, conflict reported	**Success**
SLOT-10	**Integration**	Create a slot touching an existing boundary	09:30-10:00 after 09:00-09:30	Slot created, ranges only touch	**Success**
SLOT-11	**Integration**	Update a slot without any field	Empty JSON object	Validation error shown, slot unchanged	**Success**
SLOT-12	**Integration**	Patient attempts to create a clinic slot	Authenticated patient token	Access denied, staff and admin only	**Success**
SLOT-13	**Integration**	Reduce a slot capacity below booked seats	One booked seat, set to 0	Reduction rejected, capacity unchanged	**Success**
SLOT-14	**Integration**	Delete a slot with an active appointment	Slot with one BOOKED appointment	Deletion rejected, slot retained	**Success**

SLOT-15	**Integration**	Delete a slot with no active appointments	Slot with no bookings	Slot deleted successfully	**Success**

QUEUE MANAGEMENT

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
QUEUE-01	**Unit**	Generate a queue number for a patient	Patient with confirmed appt or walk-in	Unique queue number is generated for patient	**Success**
QUEUE-02	**Integration**	Add a patient to queue after a successful booking	Valid appt and available slot	Patient added to correct queue and associated with appt	**Success**
QUEUE-03	**E2E**	Patient joins queue successfully	Registered patient with valid appt	Patient receives queue number and confirmation	**Pending**
QUEUE-04	**Unit**	Maintain the correct queue order	Multiple patients with differing positions	Patients arranged according to defined queue rules	**Fail**

QUEUE-05	**Integration**	Update a patient's queue status	Patient marked as Waiting	Status updated correctly and reflected in display	**Success**
QUEUE-06	**E2E**	Staff notify next patient	Multiple patients waiting	Next eligible patient found, status changes to Called	**Success**
QUEUE-07	**Integration**	Prevent duplicate queue entries	Patient already has active entry	System prevents unintended duplicate entry	**Pending**
QUEUE-08	**Unit**	Handle an empty queue	No patients waiting	Displays empty message, doesn't call nonexistent	**Pending**
QUEUE-09	**E2E**	Mark a patient as served	Patient called, consult completed	Status changes to Served, queue advances to next	**Pending**
QUEUE-10	**Integration**	Handle a skipped or absent patient	Patient notified but no response	Staff marks as Skipped/ No-show, queue proceeds	**Pending**
QUEUE-11	**Integration**	Synchroni ze queue updates	Staff updates status	Displays show updated info, no conflict	**Pending**

		across UIs			

NOTIFICATION MANAGEMENT

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
NOTIF-01	**E2E**	Patient receives booking confirmation	Valid patient account, slot, details	Appt booked, confirmation notification sent or displayed	**Fail**
NOTIF-02	**Integration**	Patient receives cancellation notification	Patient with existing appt, cancelled	System sends/ displays cancellation with correct details	**Fail**
NOTIF-03	**Integration**	Patient receives rescheduling notification	Existing appt, successfully resched.	System notifies patient of updated date and time	**Fail**
NOTIF-04	**Integration**	Patient receives notification when status changes	Patient with active queue, update	System sends/ displays notification reflecting latest status	**Fail**

NOTIF-05	**E2E**	Patient receives notification when number is called	Patient waiting, staff auth to call	Number called, notification delivered through channel	**Fail**
NOTIF-06	**Integration**	Prevent duplicate notifications same event	One booking or update trigger	System does not send unintended duplicate	**Fail**
NOTIF-07	**Integration**	Handle delivery failure	Valid event, unavailable	Handles failure appropriately, records/retries	**Fail**
NOTIF-08	**Unit**	Validate content	Template with details	Contains correct values, no missing fields	**Fail**
NOTIF-09	**Integration**	Staff receives relevant notifications	Auth staff, relevant event	Delivers notification without exposing other info	**Fail**
NOTIF-10	**E2E**	Patient views notifications	Auth patient, existing notifs	Patient views own notifs, info matches activity	**Fail**

NOTIF-11	**Unit**	Mark notification as read	Existing unread notif	Read status changes, no unrelated changes	**Fail**

DATA INTEGRITY & DATABASE MANAGEMENT

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
DATA-01	**Integration**	System saves patient and appt records to DB	Valid patient info, details	Records saved correctly, retrievable from DB	**Success**
DATA-02	**Integration**	Updates DB records after cancellation	Existing appt with active status	Status updated to Cancelled, reflected in DB	**Success**
DATA-03	**Integration**	Updates DB after rescheduling	Existing appt, valid new schedule	Details saved correctly, no duplicates	**Success**
DATA-04	**Integration**	Maintains consistency appts & queue	Patient with confirmed appt	Queue entry associated with patient & record	**Pending**

DATA-05	**Unit**	Validates data before saving	Missing fields, bad format	Invalid data rejected by logic	**Pending**
DATA-06	**Integration**	Prevents duplicate patient records	Existing patient records	Prevents unintended duplication	**Pending**
DATA-07	**Integration**	Preserves data on failure	Existing record, simulated failure	Avoids partial changes, preserves consistency	**Pending**
DATA-08	**Integration**	Handles DB connection failure	Simulated unavailable DB	Handles gracefully, displays error, no false success	**Pending**
DATA-09	**Integration**	Retrieves latest data	Existing record created/ updated	Retrieves latest values, not outdated info	**Pending**
DATA-10	**Integration**	Prevents unauthorized modifications	Patient attempting to modify records	Modifications rejected, original data unchanged	**Pending**
DATA-11	**Integration**	Maintains consistency	Two requests modifying same rec	Concurrency rules applied, prevents conflict	**Pending**

		simultaneous updates			
DATA-12	**Integration**	Preserves records when related cancelled	Patient with appts, history	Records consistent, history retained	**Pending**

USER & ADMINISTRATIVE MANAGEMENT

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
ADMIN-01	**E2E**	Admin views registered user accounts	Auth admin, existing records	Admin views list of users and info	**Fail**
ADMIN-02	**Integration**	Admin assigns or updates user's role	Existing user, valid role	Role updated, permissions reflect role	**Fail**
ADMIN-03	**E2E**	Admin deactivates user account	Active user, auth admin	Account deactivated, cannot access protected feats	**Fail**
ADMIN-04	**E2E**	Admin reactivates user account	Deactivated user, auth admin	Account reactivated, access restored per rules	**Fail**

ADMIN-05	**Integration**	Prevents unauthorized admin actions	Auth patient/ staff, admin action	Action rejected, records unchanged	**Fail**
ADMIN-06	**Integration**	Admin updates clinic config settings	Auth admin, valid config	Settings saved and applied to functions	**Fail**
ADMIN-07	**Unit**	Validates admin config inputs	Missing, invalid, out of range	Invalid values rejected, feedback provided	**Fail**
ADMIN-08	**E2E**	Admin searches for user account	Existing user, search term	Displays matching account or message if none match	**Fail**

DASHBOARD & REPORTING MANAGEMENT

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
DASH-01	**E2E**	Clinic staff views dashboard	Auth staff, existing records	Dashboard loads successfully, shows info per scope	**Fail**

DASH-02	**Integration**	Dashboard shows correct appt count	Known set of appts	Count matches underlying records/rules	**Fail**
DASH-03	**Integration**	Dashboard shows correct active queue count	Known set of queue entries	Active count matches rules	**Fail**
DASH-04	**Integration**	Dashboard reflects appt updates	Appt booked, cancel/ resched	Dashboard reflects change per behavior	**Fail**
DASH-05	**Integration**	Dashboard reflects queue updates	Queue entries in states	Summaries/ counts reflect current status	**Fail**
DASH-06	**E2E**	Admin views clinic stats	Auth admin, period records	Dashboard shows stats for selected period and scope	**Fail**
DASH-07	**Integration**	Filters data by date range	Appt and queue records over time	Only records matching range included	**Fail**
DASH-08	**Integration**	Generates report	Existing records, type	Report contains accurate data and totals	**Fail**

DASH-09	**E2E**	Authorized user exports report	Auth account, report, format	Report exported in supported format, data matches	**Fail**
DASH-10	**Integration**	Restricts data by role	Patient, staff, admin accounts	Each user only accesses permitted data	**Pending**